
CIS437 - Cloud Computing

– Google Cloud Shell & gcloud CLI –

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Last Class

- From centralized computing to cloud computing
- Virtualization and containers
- What is a cloud platform?
- Google Cloud services and interfaces
- Cloud platform trade-offs

Agenda

- What is Cloud Shell?
- Linux command review
- **gcloud** CLI
- Manage a VM from the command line
- Cloud Storage from the command line
- Hands-on practice

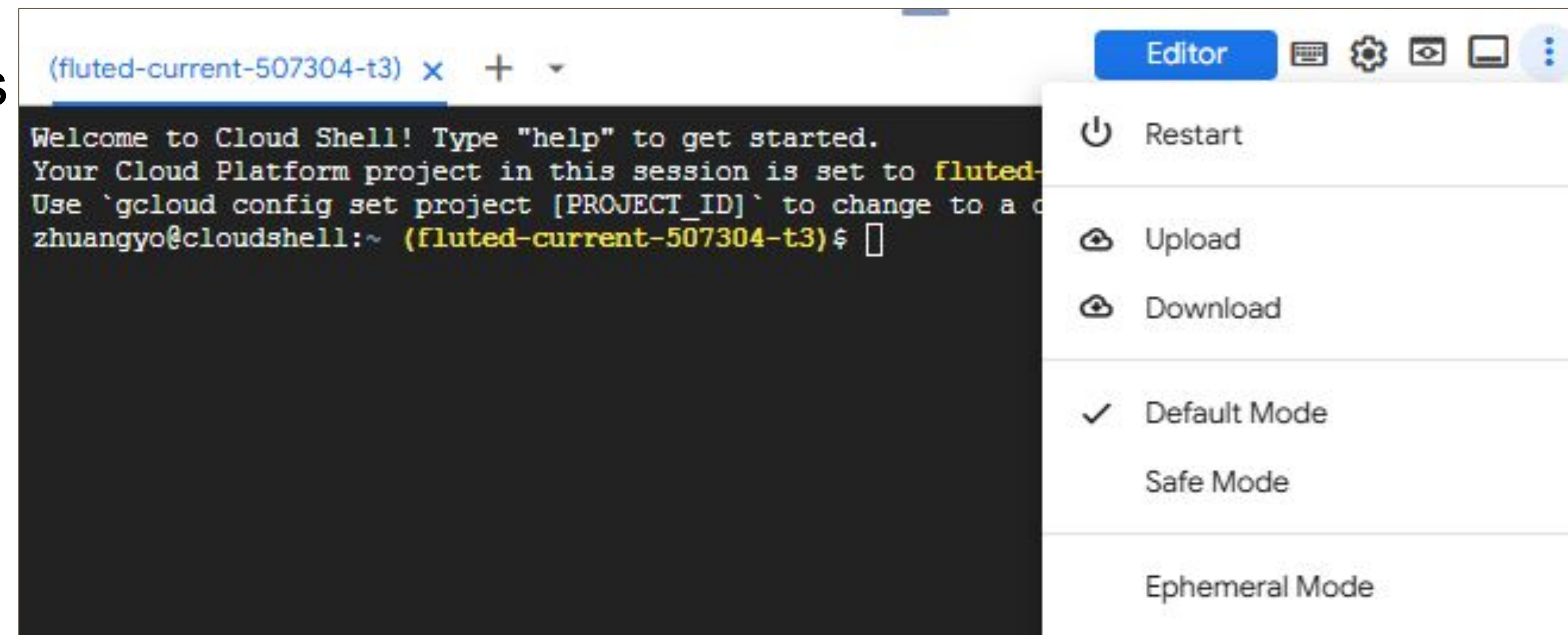
What Is Google Cloud Shell?

- Browser-based Linux shell
- Google Cloud CLI already installed
- Authenticated with your Google account
- Connected to your active Google Cloud project
- Useful for administration, scripting, and development

[Cloud console](#)

How Cloud Shell Works

- Temporary Linux VM managed by Google
- **gcloud**, Git, Python, Docker, and other tools preinstalled
- 5 GB persistent \$HOME storage by default
- Active Google Cloud project is passed into the session
- Optional **Ephemeral mode**
 - faster startup
 - no persistent home disk
 - files disappear when the session ends



Official Doc: [how cloud shell works](#)

Default mode vs. Ephemeral mode

Default Cloud Shell	Ephemeral mode
Temporary VM	Temporary VM
5 GB persistent \$HOME	No persistent home disk
Files survive sessions	Files deleted after session
Slightly more setup	Faster startup

What Can You Do with Cloud Shell?

- Browser-based Linux environment
- Manage Google Cloud resources with `gcloud`
- Use common tools such as Git, Python, and text editors
- Run scripts and automation
- Access the active Google Cloud project

Today's Focus: The Interface, Not Every Service

- We will touch several Google Cloud services
- Some services will be covered in detail later
- Focus today on using **Cloud Shell** and **gcloud**
- Don't worry about mastering every service yet

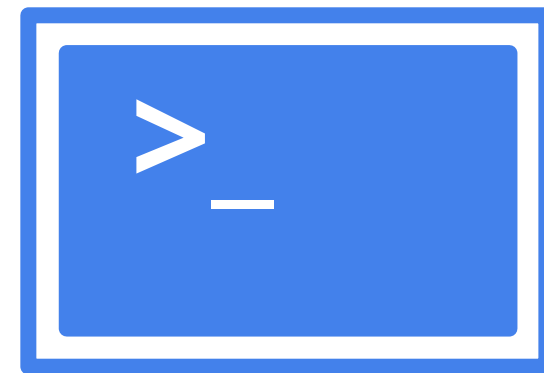
Ways to Interact with Google Cloud

There are four ways to interact with Google Cloud



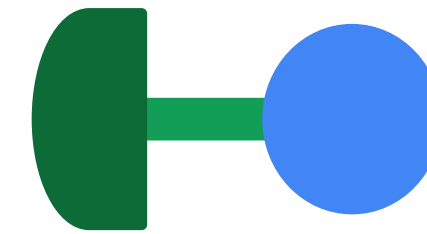
**Google Cloud
Console**

Web user
interface



**Cloud SDK and
Cloud Shell**

Command-line
interface



**REST-based
API**

For custom
applications



**Cloud Console
mobile app**

For iOS and
Android

Example 1: Basic usage

File management (i.e., common commands)

- show current directory: `pwd`
- list files: `ls`
- change directory: `cd directory`
- create directory: `mkdir directory`
- move file: `mv file1 file2`
- copy file: `cp file1 file2`
- copy directory: `cp -r dir1
dest_directory`
- delete file: `rm file`
- delete directory: `rm -rf dir1`



Your default Cloud Shell home directory has 5GB persistent storage.

Be careful with `rm -rf`. It deletes recursively and does not ask for confirmation.

Git

- clone: `git clone <repository url>`

[Cloud console](#)

Common GCP commands

Google Cloud / gcloud

- show current configuration: `gcloud config list`
- show active project: `gcloud config get-value project`
- set active project: `gcloud config set project PROJECT_ID`
- set default region: `gcloud config set compute/region us-east1`
- show active account: `gcloud auth list`
- list VM instances: `gcloud compute instances list`
- get help: `gcloud --help`
- get help for a command: `gcloud compute --help`

Shell variables

- set variable: `myvar="HELLO THERE"`
- show variable: `echo $myvar`

First Things to Check in Cloud Shell

Open Cloud Shell



Cloud Shell is usually already authenticated



Check active account: `gcloud auth list`



Check active project: `gcloud config get-value project`



Wrong project?: `gcloud config set project PROJECT_ID`



Check current configuration if needed: `gcloud config list`



Run your gcloud command

Example 2: Create a VM with gcloud

- Same VM concepts as the Console
- Need to choose:
 - VM name
 - zone
 - machine type
 - OS image
- We will use a public Ubuntu image

Cloud console

Example 2: Create a VM with gcloud

```
$ gcloud compute images list
```

```
NAME: rocky-linux-8-optimized-gcp-v20260910
PROJECT: rocky-linux-cloud
FAMILY: rocky-linux-8-optimized-gcp
DEPRECATED:
STATUS: READY

NAME: rocky-linux-8-v20260910
PROJECT: rocky-linux-cloud
FAMILY: rocky-linux-8
DEPRECATED:
STATUS: READY

NAME: rocky-linux-9-arm64-v20260910
PROJECT: rocky-linux-cloud
FAMILY: rocky-linux-9-arm64
DEPRECATED:
STATUS: READY

NAME: rocky-linux-9-optimized-gcp-arm64-v20260910
PROJECT: rocky-linux-cloud
FAMILY: rocky-linux-9-optimized-gcp-arm64
DEPRECATED:
STATUS: READY
```

```
$ gcloud compute images describe-from-family
ubuntu-2404-lts-amd64 \
  --project=ubuntu-os-cloud
```

```
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud compute images describe-from-family ubuntu-2404-lts-amd64
--project=ubuntu-os-cloud
architecture: X86_64
archiveSizeBytes: '68223366400'
creationTimestamp: '2026-09-06T03:15:26.996-07:00'
description: Canonical, Ubuntu, 24.04 LTS, amd64 noble image built on 2026-09-06
diskSizeGb: '10'
enableConfidentialCompute: false
family: ubuntu-2404-lts-amd64
```

Official Doc: [Create and start a Compute Engine instance](#)

Zones and machine types

```
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud compute zones list
```

```
NAME: us-east1-b  
REGION: us-east1  
STATUS: UP  
NEXT_MAINTENANCE:  
TURNDOWN_DATE:
```

```
NAME: us-east1-c  
REGION: us-east1  
STATUS: UP  
NEXT_MAINTENANCE:  
TURNDOWN_DATE:
```

```
NAME: us-east1-d  
REGION: us-east1  
STATUS: UP  
NEXT_MAINTENANCE:  
TURNDOWN_DATE:
```

First, what zones are there?

```
$ gcloud compute zones list
```

When a command returns too much output, either pipe it to `| less` or use `--filter` to narrow the result. Use `Ctrl + L` to clear the terminal screen

```
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud compute machine-types list \
```

```
--zones=us-east1-b \  
--filter="name=e2-micro"
```

```
NAME: e2-micro  
ZONE: us-east1-b  
CPUS: 2  
MEMORY_GB: 1.00  
DEPRECATED:
```

And what machine types in that zone?

```
$ gcloud compute machine-types list --zones us-east1-b
```

Filter for a specific type:

```
$ gcloud compute machine-types list \  
--zones=us-east1-b \  
--filter="name=e2-micro"
```

Create the VM

```
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud compute instances create shell-demo-vm \
--zone=us-east1-b \
--machine-type=e2-micro \
--image-family=ubuntu-2404-lts-amd64 \
--image-project=ubuntu-os-cloud
Created [https://www.googleapis.com/compute/v1/projects/fluted-current-507304-t3/zones/us-east1-b/instances/shell-demo-vm].
WARNING: Some requests generated warnings:
- You are creating a global DNS VM. VM instances using global DNS names may experience service disruptions during network outages. To reduce the risk of widespread service disruption, please refer to the documentation at https://cloud.google.com/compute/docs/networking/zonal-dns-vm.

NAME: shell-demo-vm
ZONE: us-east1-b
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.3
EXTERNAL_IP: 34.75.84.27
STATUS: RUNNING
```

```
$ gcloud compute instances create VM_NAME \
--zone=ZONE \
[--image=IMAGE | --image-family=IMAGE_FAMILY] \
--image-project=IMAGE_PROJECT \
--machine-type=MACHINE_TYPE
```

i.e.,

```
$ gcloud compute instances create shell-demo-vm \
--zone=us-east1-b \
--machine-type=e2-micro \
--image-family=ubuntu-2404-lts-amd64 \
--image-project=ubuntu-os-cloud
```

Verify the VM

```
$ gcloud compute instances list
```

```
NAME: shell-demo-vm
ZONE: us-east1-b
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.142.0.3
EXTERNAL_IP: 34.75.84.27
STATUS: RUNNING
```

Same resource, different interface.

VM instances							
 Filter Enter property name or value							
☐ Status	Name 	Zone	Recommendations	In use by	Internal IP	External IP	Connect
☐ 	shell-demo-vm	us-east1-b			10.142.0.3 (nic0)	34.75.84.27 (nic0)	SSH 
☐ 	zhuang-cc-microinstance-1	us-central1-a			10.128.0.5 (nic0)		SSH 

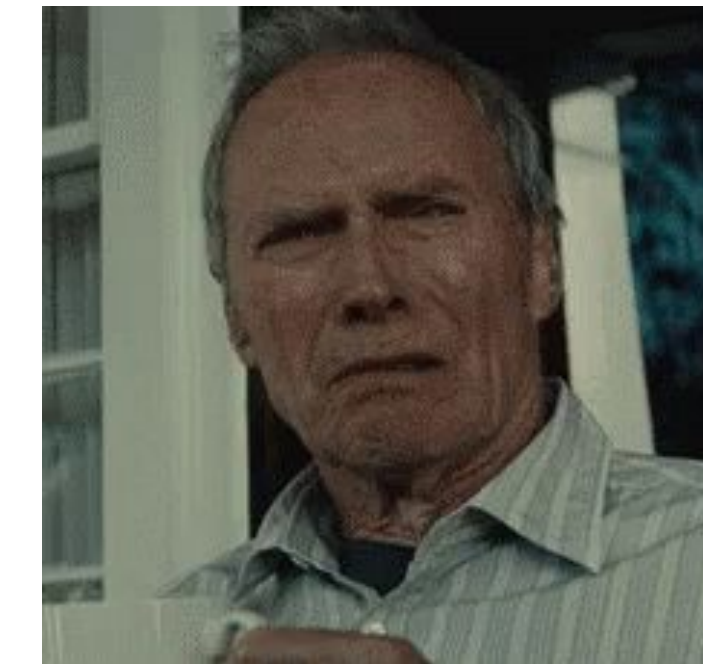
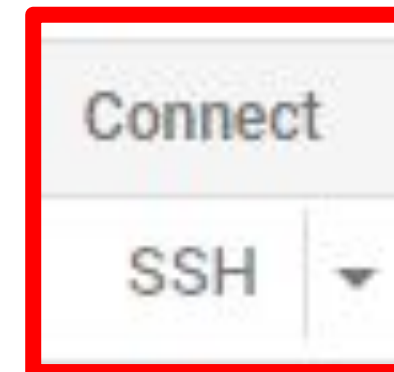
Remember: Cloud Resources Can Cost Money

- Running VM compute can consume credits
- Stopping a VM stops compute charges
- Persistent disks and some other resources may remain billable
- Delete temporary demo resources when finished

☐	Status	Name ↑	Zone	Recommendations
☐		shell-demo-vm	us-east1-b	

SSH to the VM

We don't need this pesky thing anymore!



```
zhuangyo@cloudshell:~ (fluted-current-507304-t3)$ gcloud compute ssh shell-demo-vm --zone=us-east1-b
Existing host keys found in /home/zhuangyo/.ssh/google_compute_known_hosts
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-1011-gcp x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Tue Sep 15 14:42:16 UTC 2026

System load:  0.0                Processes:           115
Usage of /:   30.1% of 8.65GB    Users logged in:    0
Memory usage: 29%                IPv4 address for ens4: 10.142.0.3
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

14 updates can be applied immediately.
8 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Sep 15 14:38:37 2026 from 35.231.194.133
```

Cloud Shell



`gcloud compute ssh`



Compute Engine VM

We have the Cloud Shell!

```
$ gcloud compute ssh <vm> --zone=<zone>
```

also! allows for tab completion!

i.e.,

```
$ gcloud compute ssh shell-demo-vm --zone=us-east1-b
```

Cloud Shell Editor

- Browser-based code editor built into Cloud Shell
- Based on Code OSS
- Edit the same files stored in your Cloud Shell environment
- Integrated terminal
- Git and debugging support
- Useful for developing and testing cloud applications

Official Doc: [Launch the Cloud Shell Editor](#)

Cloud Shell Editor

From the Cloud Shell terminal, you can open the editor with:

```
$ cloudshell edit .
```

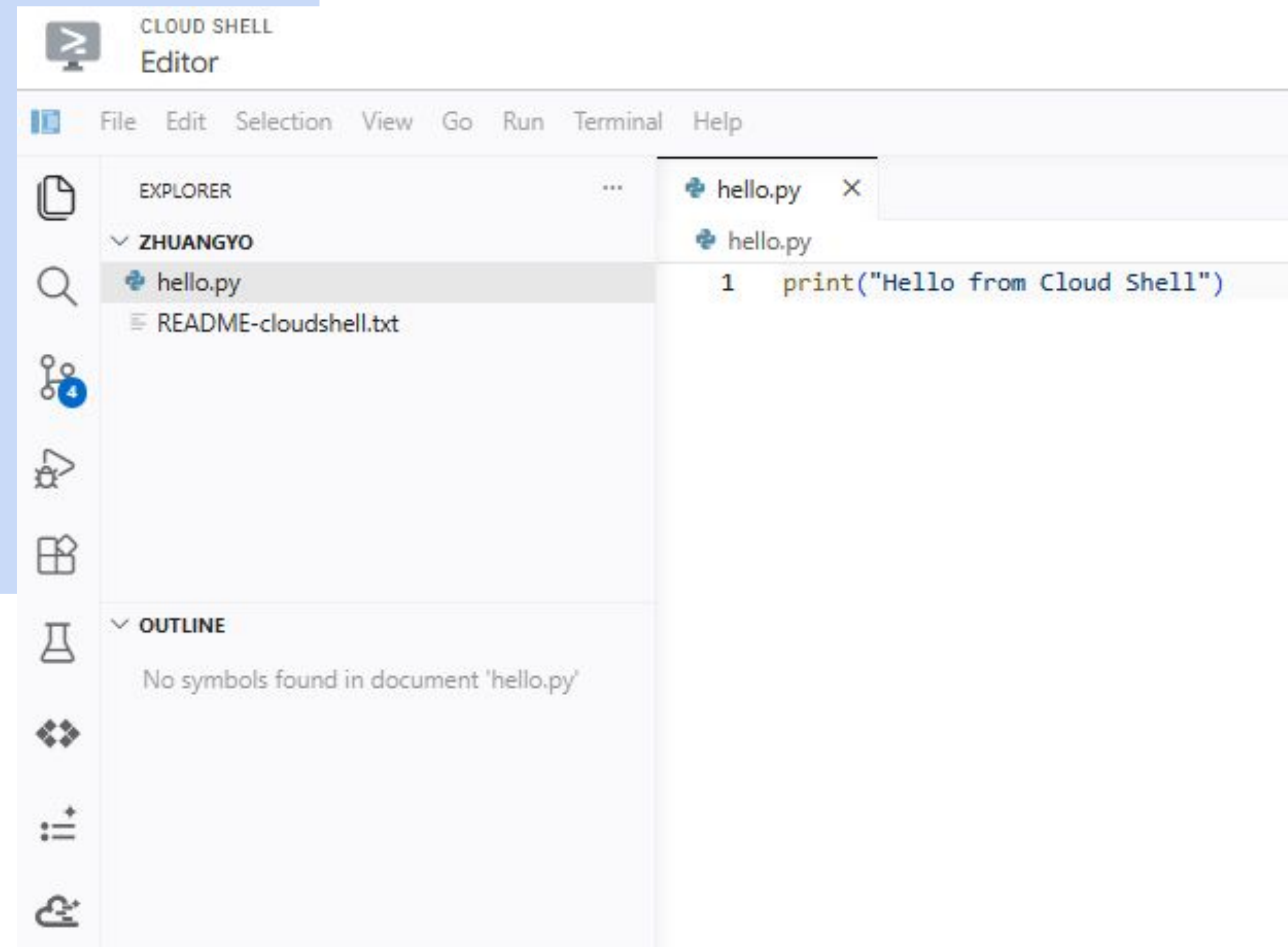
Then create a small file in the editor, for example:

```
print("Hello from Cloud Shell")
```

Save it as: [hello.py](#)

Then in the terminal:

```
python3 hello.py
```



Running gcloud on Your Own Computer

- Google Cloud CLI can be installed locally
- Available for Windows, macOS, and Linux
- Same **gcloud** commands as Cloud Shell
- Requires local installation and authentication
- Useful for development scripts and automation

Official Doc: [Install the Google Cloud CL](#) | [Use Cloud Shell with the gcloud CLI](#)

First-Time Setup on a Local Computer

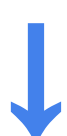
Install Google Cloud CLI



Run: `$gcloud init`



Sign in with Google account



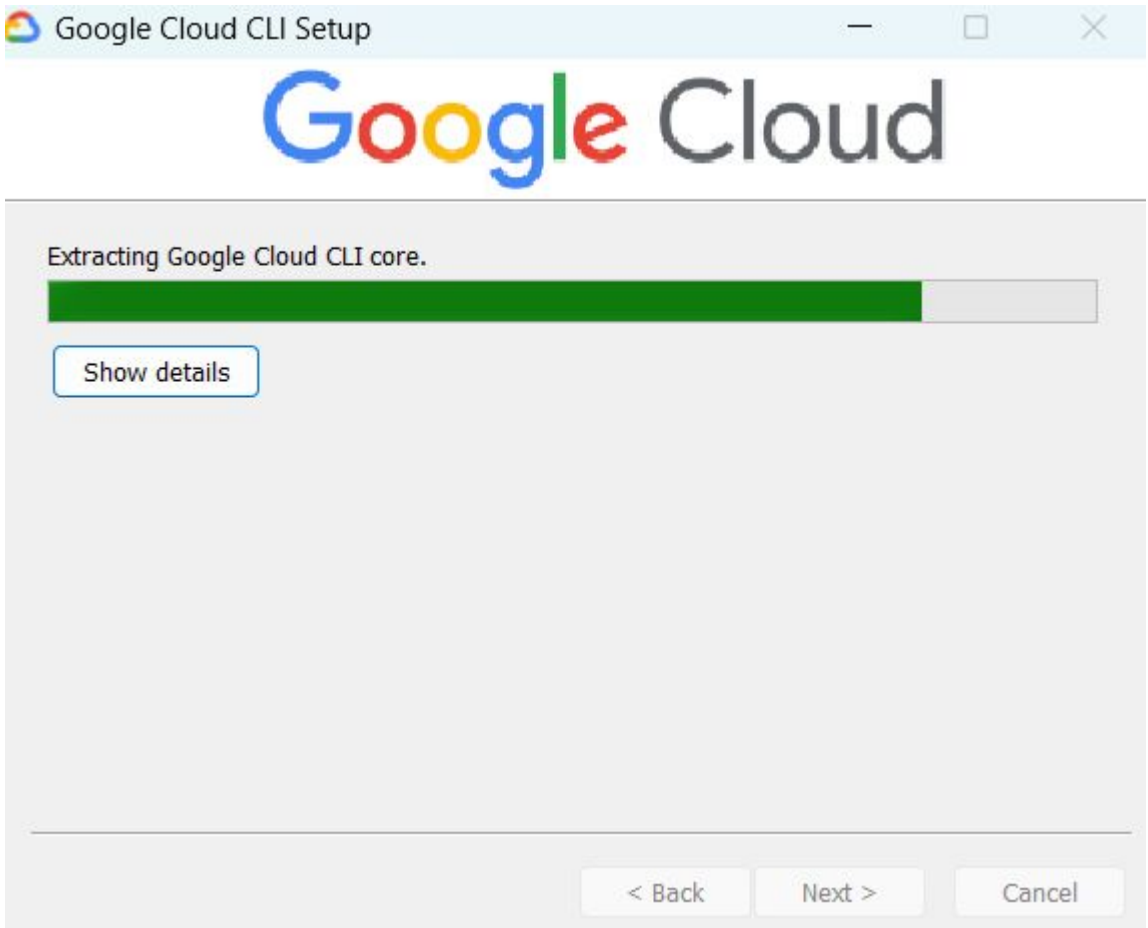
Choose a Google Cloud project



Optional: configure defaults



Ready to use gcloud



Home > Documentation > Developer tools > Google Cloud SDK

Was this helpful?

You are now authenticated with the gcloud CLI!

Send feedback

The authentication flow has completed successfully. You may close this window, or check out the resources below.

Information about command-line tools and client libraries

To learn more about Google Cloud CLI commands, see the [gcloud CLI guide](#).

To learn more about the command-line tools for App Engine, Compute Engine, Cloud Storage, BigQuery, Cloud SQL, and Cloud DNS (which are all bundled with the gcloud CLI), see [Accessing services with the gcloud CLI](#).

```
C:\Users\zhuangyo\AppData\Local\Google\Cloud SDK>gcloud compute instances list
NAME                                ZONE          MACHINE_TYPE  PREEMPTIBLE  INTERNAL_IP  EXTERNAL_IP  STATUS
zhuang-cc-microinstance-1         us-central1-a  e2-micro      10.128.0.5    34.75.84.27  TERMINATED
shell-demo-vm                     us-east1-b     e2-micro      10.142.0.3    34.75.84.27  RUNNING
```

Same gcloud Command, Different Environment

```
$ gcloud compute instances create ubuntu-test-vm-local --zone=us-east1-b  
--image-family=ubuntu-2404-lts-amd64 --image-project=ubuntu-os-cloud --machine-type=e2-micro
```

```
C:\Users\zhuangyo\AppData\Local\Google\Cloud SDK>gcloud compute instances create ubuntu-test-vm-local --zone=us-east1-b  
--image-family=ubuntu-2404-lts-amd64 --image-project=ubuntu-os-cloud --machine-type=e2-micro  
Created [https://www.googleapis.com/compute/v1/projects/fluted-current-507304-t3/zones/us-east1-b/instances/ubuntu-test-  
vm-local].  
WARNING: Some requests generated warnings:  
- You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross-regional outages. To reduce t  
he risk of widespread service disruption, use zonal DNS instead. Learn more at https://cloud.google.com/compute/docs/net  
working/zonal-dns?utm_id=vm-insert
```

NAME	ZONE	MACHINE_TYPE	PREEMPTIBLE	INTERNAL_IP	EXTERNAL_IP	STATUS
ubuntu-test-vm-local	us-east1-b	e2-micro		10.142.0.5	35.237.148.109	RUNNING

*Cloud Shell is convenient, but **gcloud** itself is portable. You can run the same Google Cloud CLI from Cloud Shell or from your own computer.*

Example 3: Creating and filling a Bucket

What is a bucket?

- Object storage container
- Different from a VM disk

Create a bucket

```
$ gcloud storage buckets create gs://UNIQUE_BUCKET_NAME \  
--location=us-east1
```

i.e.,

```
$ gcloud storage buckets create gs://zhuang-demo-20260915 --location=us-east1
```

*Bucket name must be **globally unique***

Example 3: Creating and filling a Bucket

Create/upload a file

```
echo "Hello CIS437" > hello.txt
```

```
gcloud storage cp hello.txt gs://zhuang-demo-20260915
```

Verify and clean up

```
gcloud storage ls gs://zhuang-demo-20260915
```

```
gcloud storage rm gs://zhuang-demo-20260915/hello.txt
```

```
gcloud storage buckets delete gs://zhuang-demo-20260915
```